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Section B

Answer **all** *questions in the spaces provided.*

Use the information in the separate Resource Folder to answer the following questions.

6. *(a)* Use the information in **Table 1** to answer the following question.

Tick (✓) the boxes next to the **three** correct statements. [3]

Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto	☐
Mercury is the hottest planet	☐
All the rocky planets have an atmosphere containing carbon dioxide	☐
Mars and the Earth have the same day length	☐
The Earth has the greatest density	☐
The Moon is the smallest planet	☐

(b) Use the information in **Diagram 3** and your knowledge to answer the following questions.

(i) State how the wavelength of the dark line, λ, for the very distant galaxy is different from the same line in the nearby galaxy. [1]

....

(ii) State how the frequency of this line for a nearby galaxy compares with the same line in a very distant galaxy. [1]

....

(iii) State **one** difference about the motion of the very distant galaxy compared to the nearby galaxy. [1]

....

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(c) Use the information in **Table 1** to answer the following questions.

(i) Europa is an asteroid. Estimate its temperature and orbital period. [2]

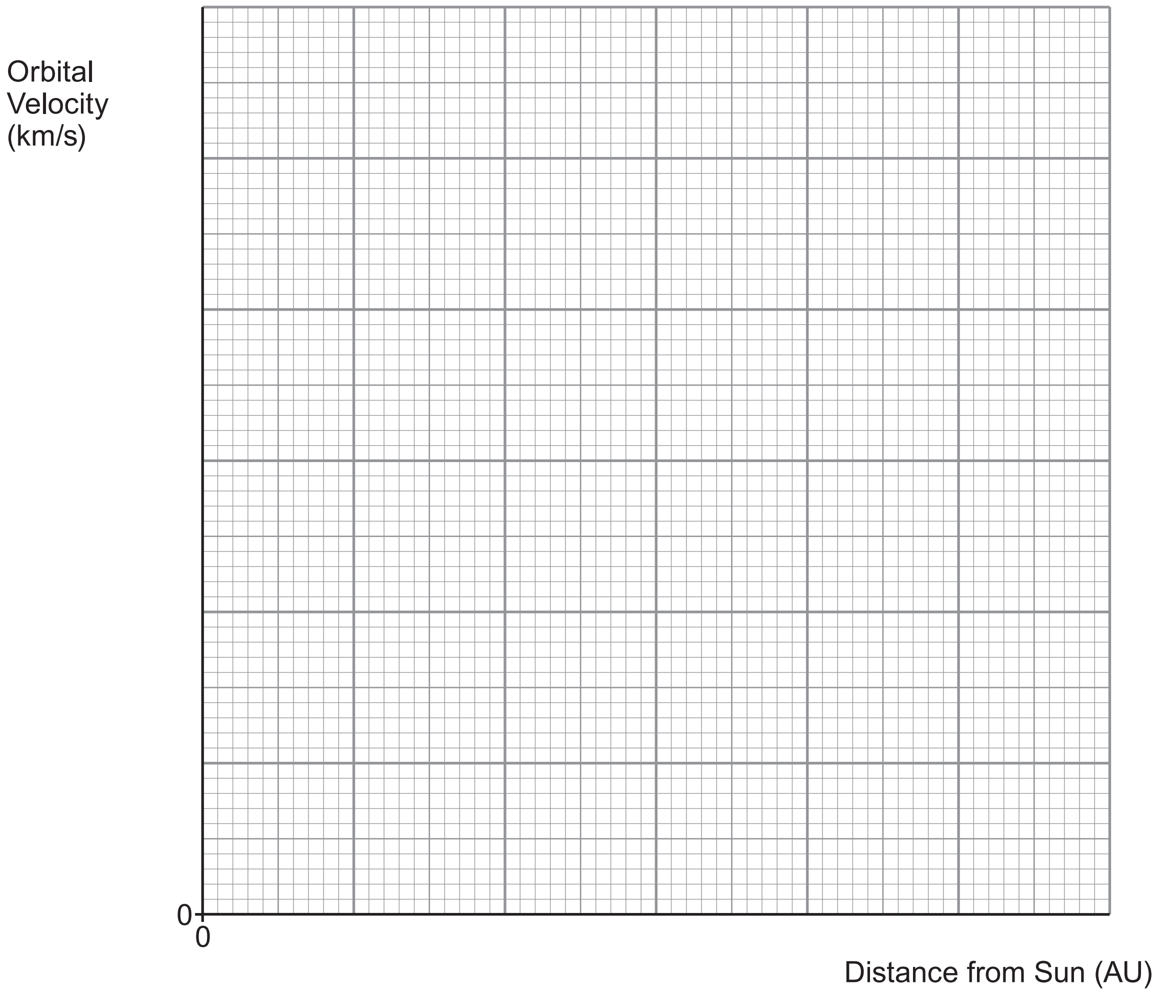
Temperature = ... °C

Orbital period = ... days

(ii) Pluto is no longer classed as a planet. State **one** reason why. [1]

....

(iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line. [4]



(iv) Orbital velocity is not proportional to the distance from the Sun.
Explain how your graph shows this to be true. [2]

....

....

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(d) Use your knowledge and the information on **pages 4** and **5** to answer the following questions.

(i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

....

....

(ii) State **two** ways in which human activities may be affected by solar flares. [2]

1. ....

2. ....

(iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

....

....

....

(e) Use the information in **Diagram 1** to describe the differences between Aristotle's model of the Solar System and the 2006 model. [4]

....

....

....

....

....

....

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....

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END OF PAPER

